



Efficient Estimation of Word Representations in Vector Space

Word2Vec? and Why?

Example:

The kid said he would grow up to be superman

The child said he would grow up to be superman

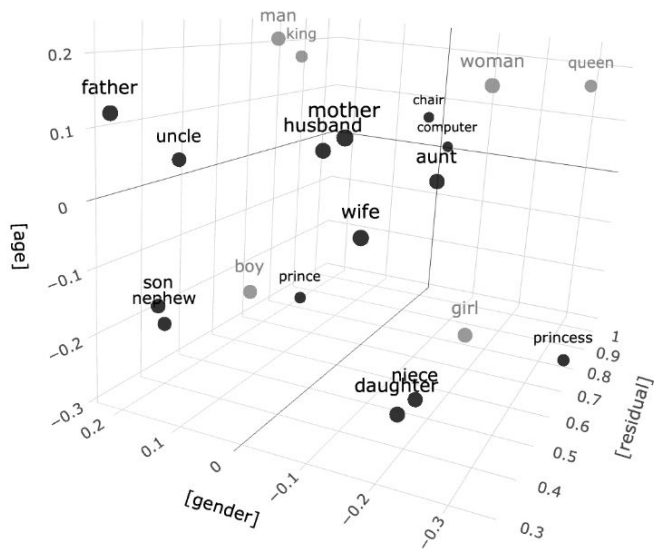
Word Embedding Intuition

Attribute	Battle	Horse	King	Man	Queen	Woman
Authority	0.8	0.2	1	0.6	1	0.5
Event	1	0	0	0	0	0
Has Tail?	0	1	0	0	0	0
Rich	0.1	0.5	1	0.4	1	0.3
Gender	0	1	-1	-1	1	1

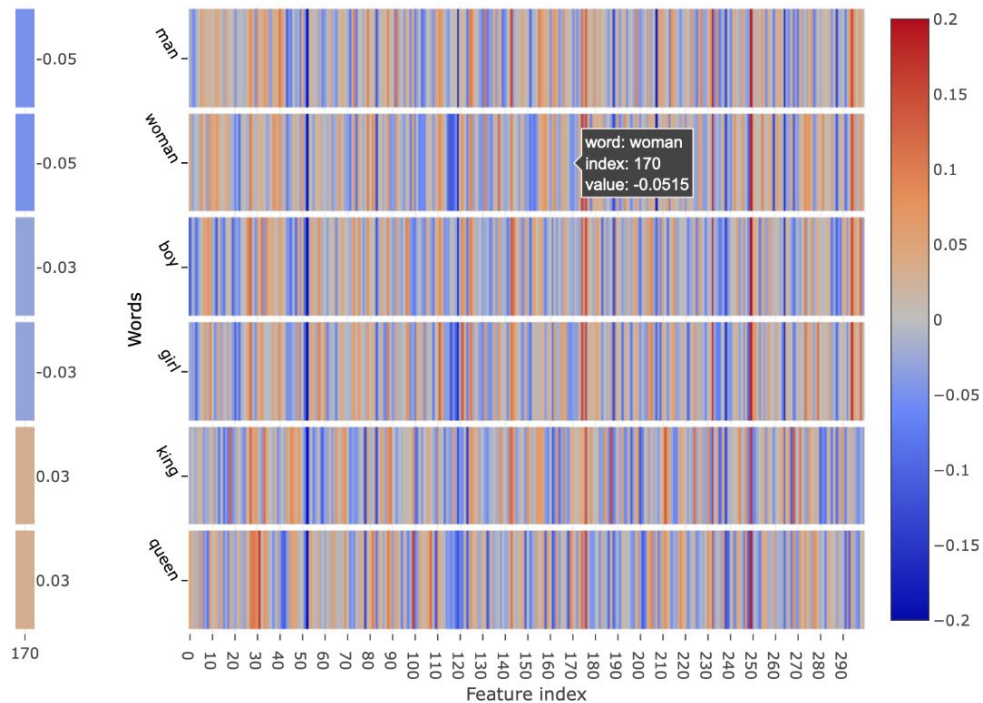
$$\begin{array}{|c|} \hline \text{King} \\ \hline 1 \\ \hline 0 \\ \hline 0 \\ \hline 1 \\ \hline -1 \\ \hline \end{array}
 -
 \begin{array}{|c|} \hline \text{Man} \\ \hline 0.6 \\ \hline 0 \\ \hline 0 \\ \hline 0.4 \\ \hline -1 \\ \hline \end{array}
 +
 \begin{array}{|c|} \hline \text{Woman} \\ \hline 0.5 \\ \hline 0 \\ \hline 0 \\ \hline 0.3 \\ \hline 1 \\ \hline \end{array}
 =
 \begin{array}{|c|} \hline 0.9 \\ \hline 0 \\ \hline 0 \\ \hline 0.9 \\ \hline 1 \\ \hline \end{array}
 \approx
 \begin{array}{|c|} \hline \text{Queen} \\ \hline 1 \\ \hline 0 \\ \hline 0 \\ \hline 1 \\ \hline 1 \\ \hline \end{array}$$

Word Embedding Intuition

Word Vector Projection



Vector Visualization



Reference: <https://www.cs.cmu.edu/~dst/WordEmbeddingDemo/>

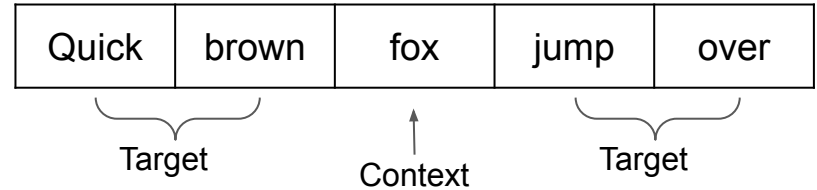
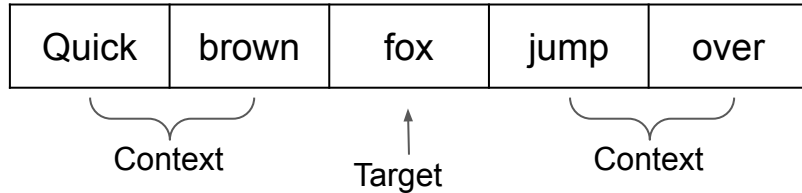
1) CBOW: Continuous Bag Of Words

- ❖ Given context words predict target word

2) Skip Gram

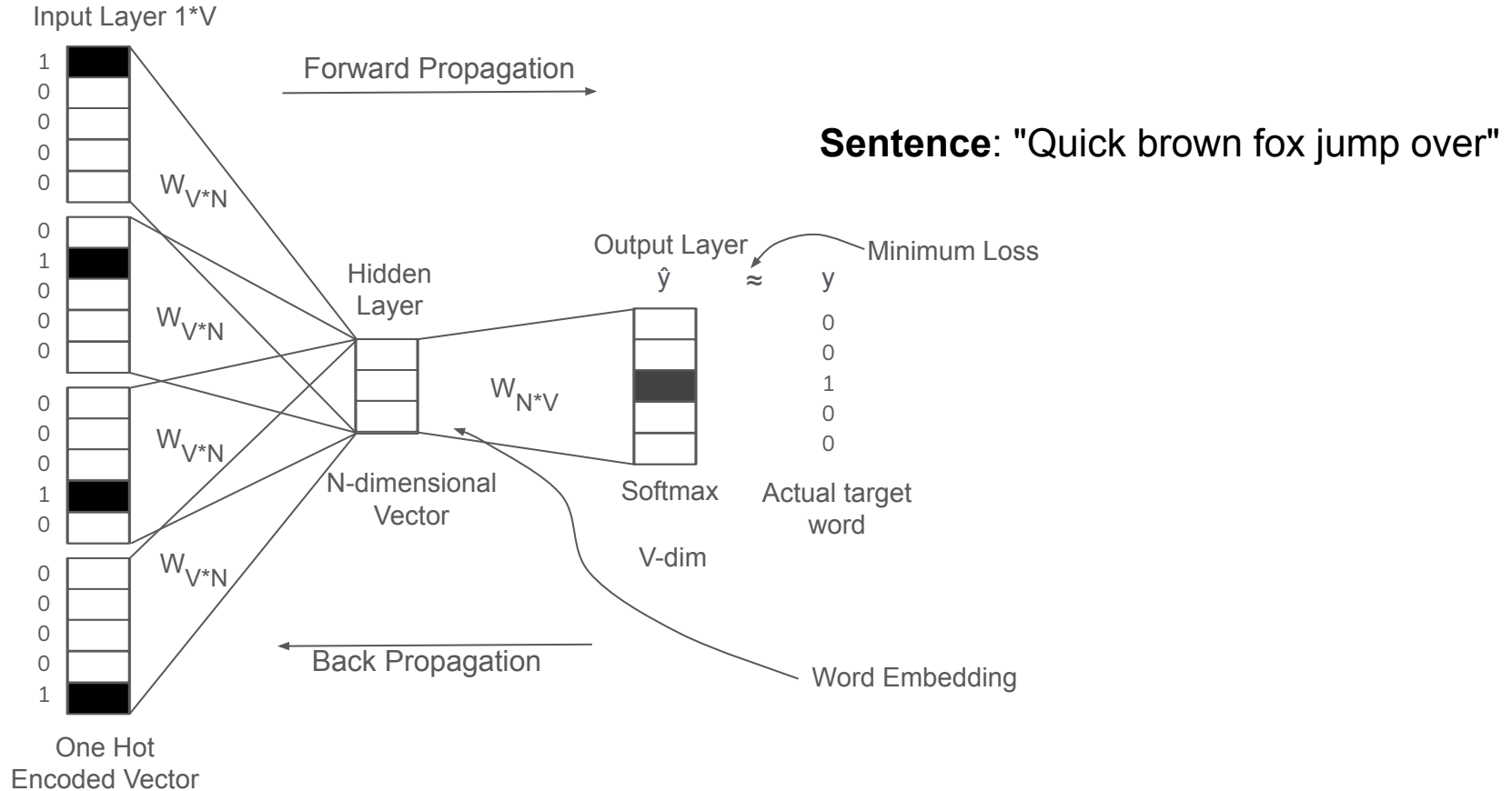
- ❖ Given the target predict context words

Sentence: "Quick brown fox jump over"



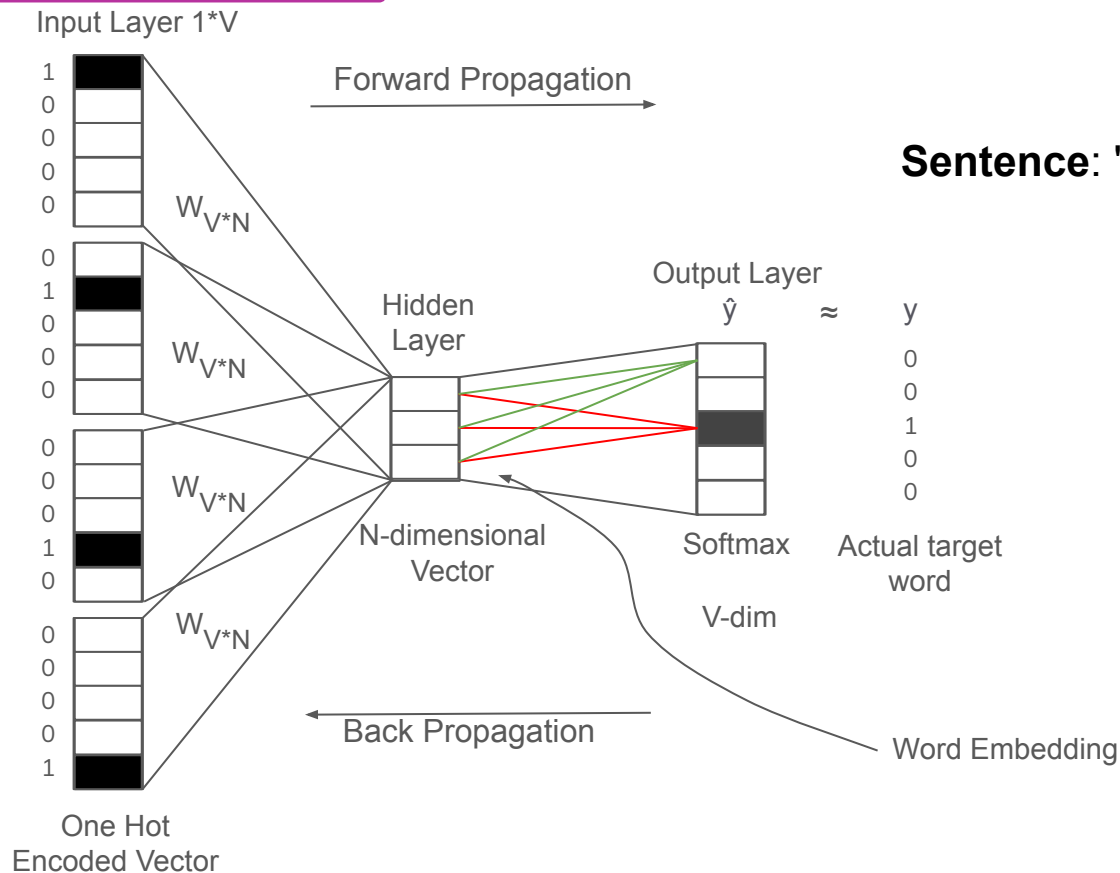
Model Architecture

CBOW: Continuous Bag of Words



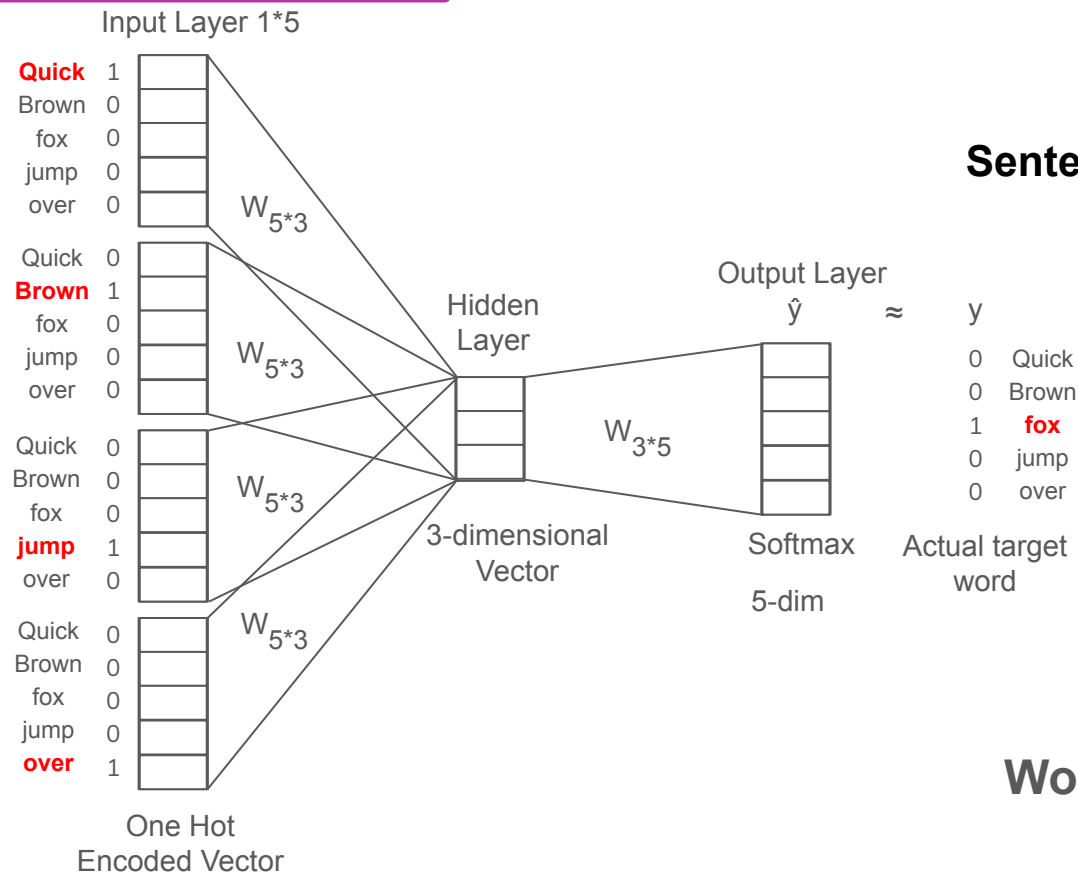
Model Architecture

CBOW: Continuous Bag of Words



Model Architecture

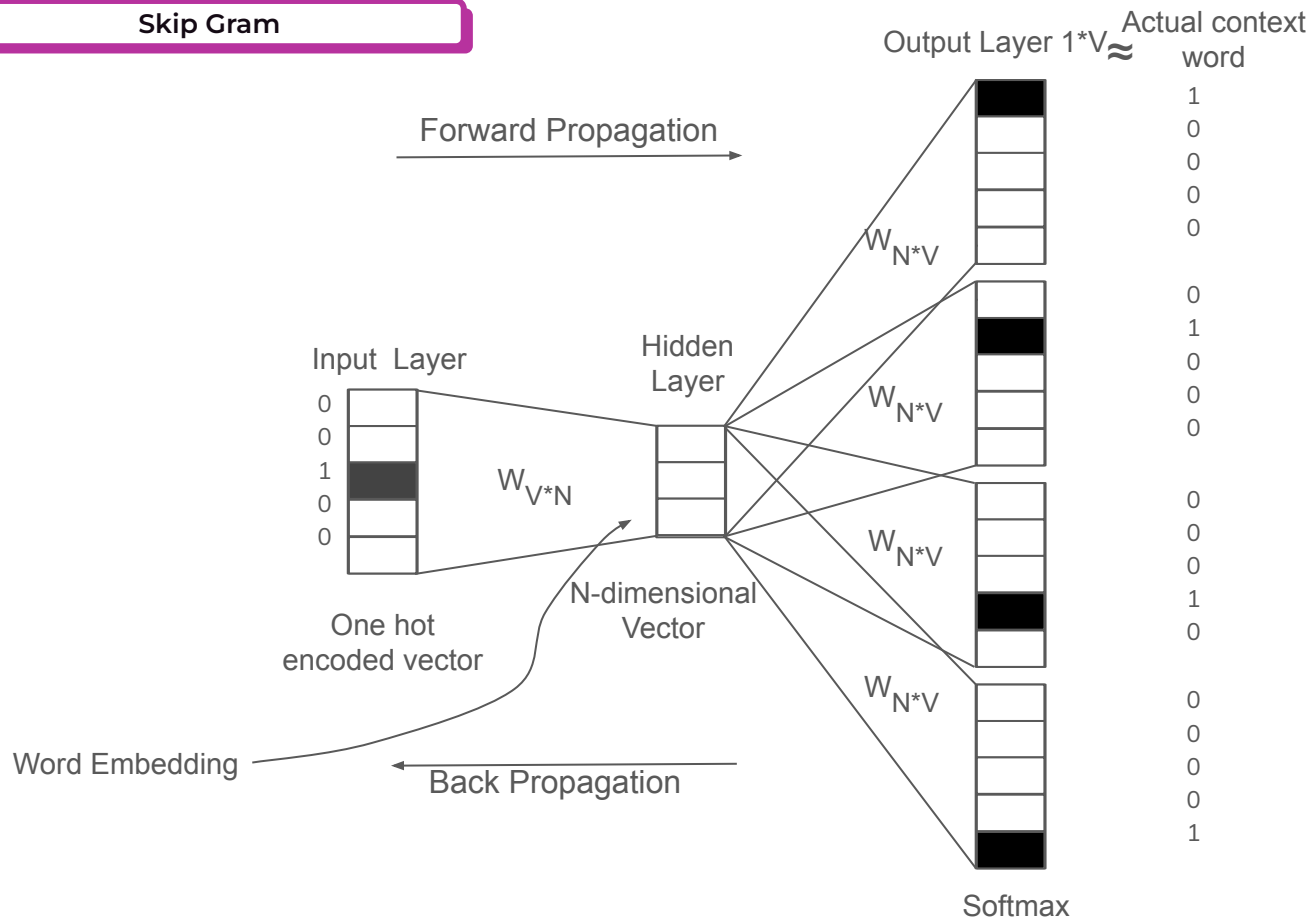
CBOW: Continuous Bag of Words



Word fox = [0.5, 0.3, 0.01]

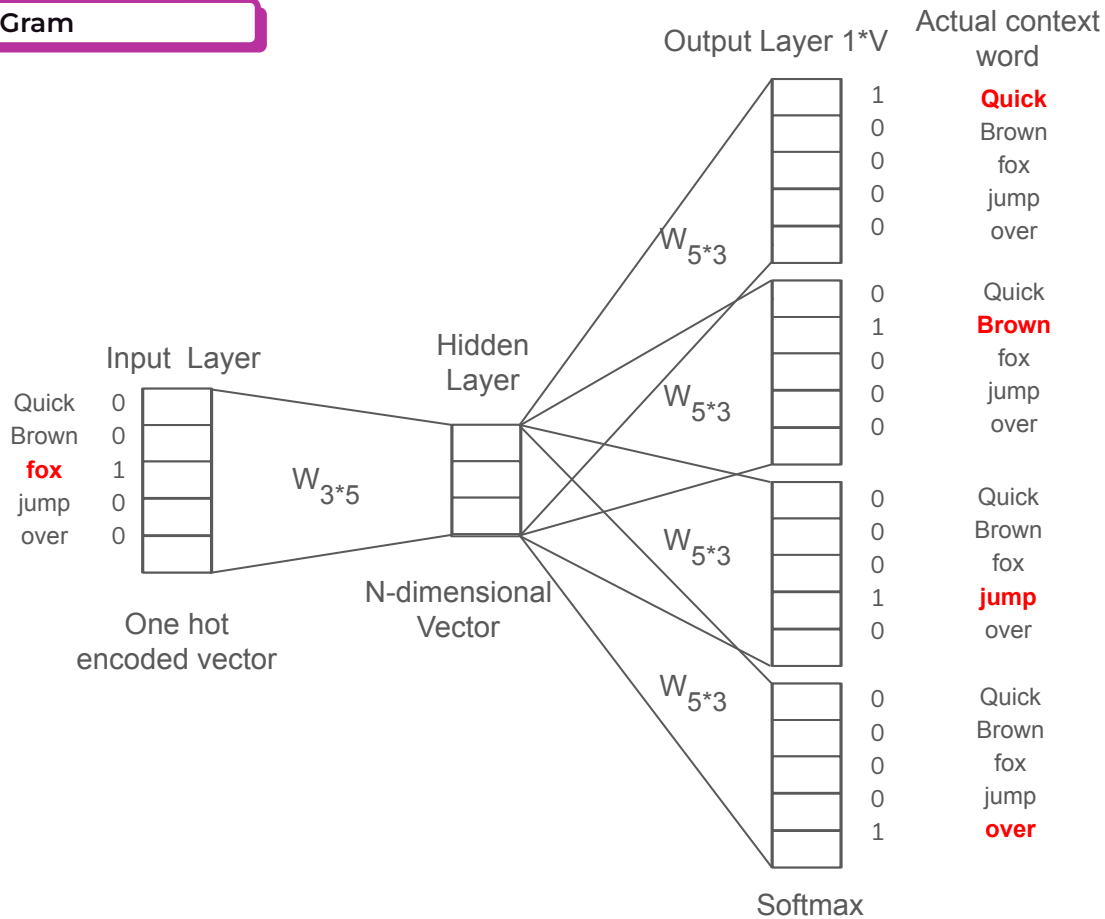
Model Architecture

Skip Gram



Model Architecture

Skip Gram



Word2Vec: Google news corpus data was used to train word vectors
6 Billion tokens (restricted to 1B most frequent words)

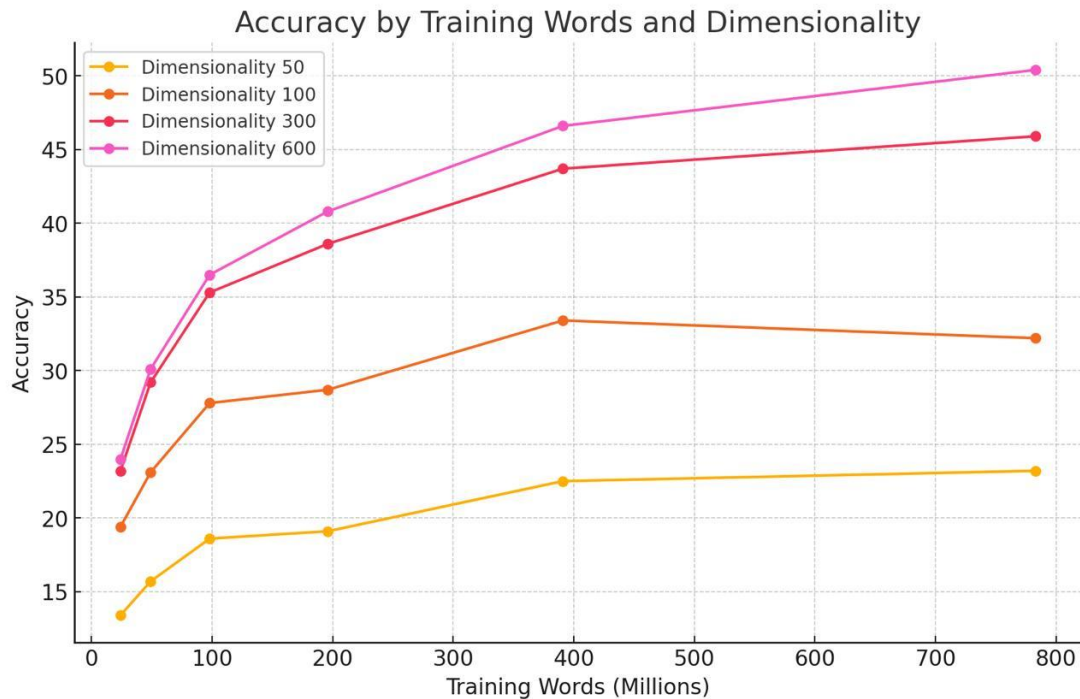


Fig1: Subset of training data with vocabulary restricted to 30k most frequent words

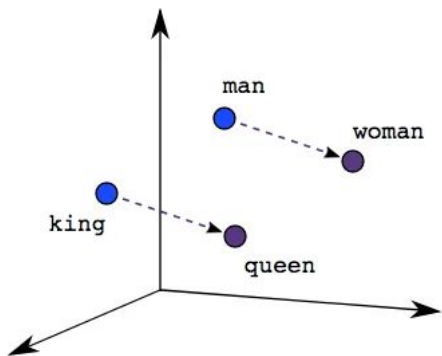
Model Architecture	Semantic-Syntactic Word Relationship test set		MSR Word Relatedness Test Set [20]
	Semantic Accuracy [%]	Syntactic Accuracy [%]	
RNNLM	9	36	35
NNLM	23	53	47
CBOW	24	64	61
Skip-gram	55	59	56

Reference: [Efficient Estimation of Word Representation in Vector Space](#)

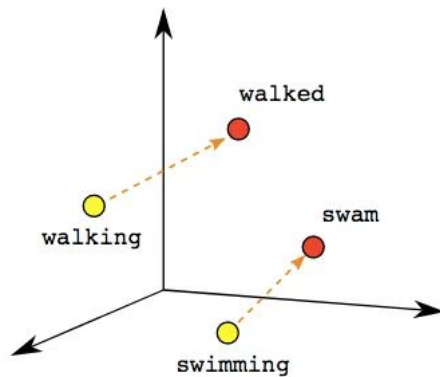
Fig2: Comparison of architectures using models trained on the same data, with 640-dimensional word vectors

- High quality representation of words into vectors
- Simple architecture and scalable
- Lower computational complexity
- Captures syntactic and semantic meanings

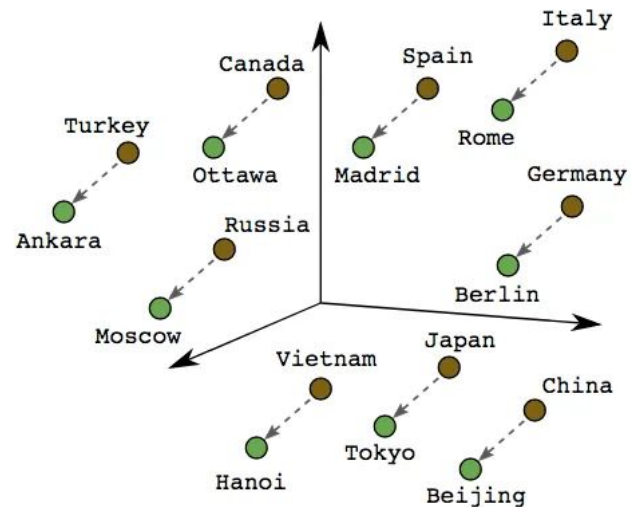
THANK YOU



Male-Female



Verb Tense



Country-Capital

Reference: <https://towardsdatascience.com/a-guide-to-word-embeddings-8a23817ab60f>